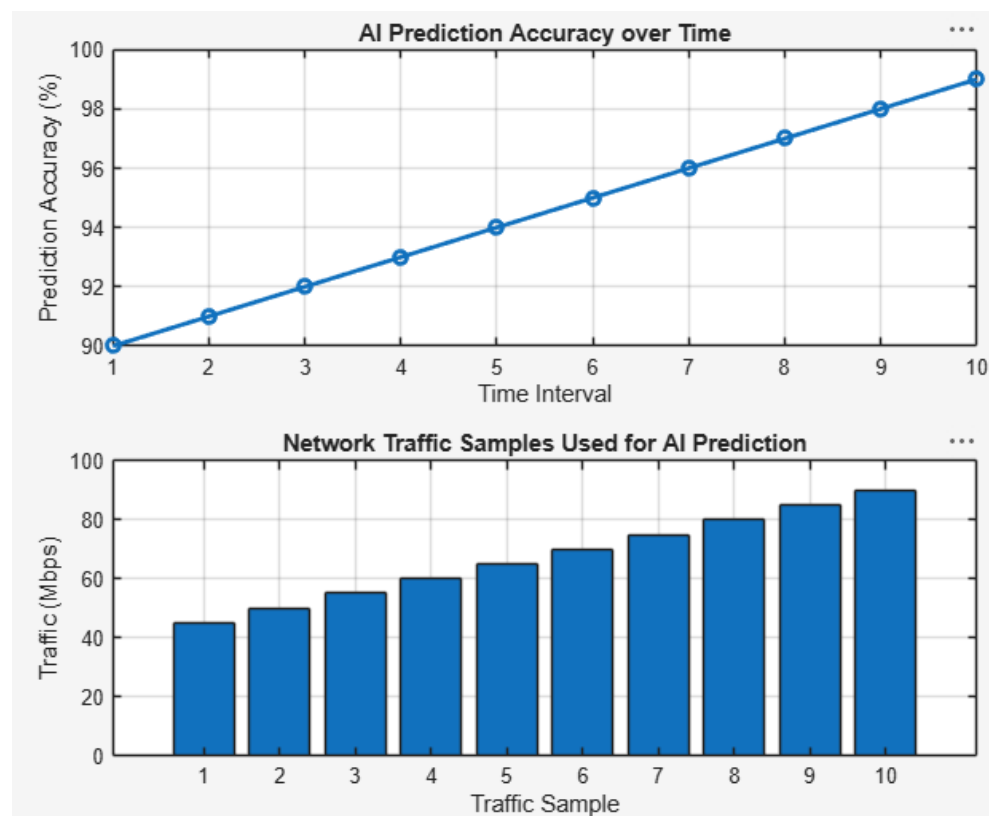


Ex.No.15: AI-based Traffic Prediction for 5G Networks

Objective 1: Matlab Code and Output

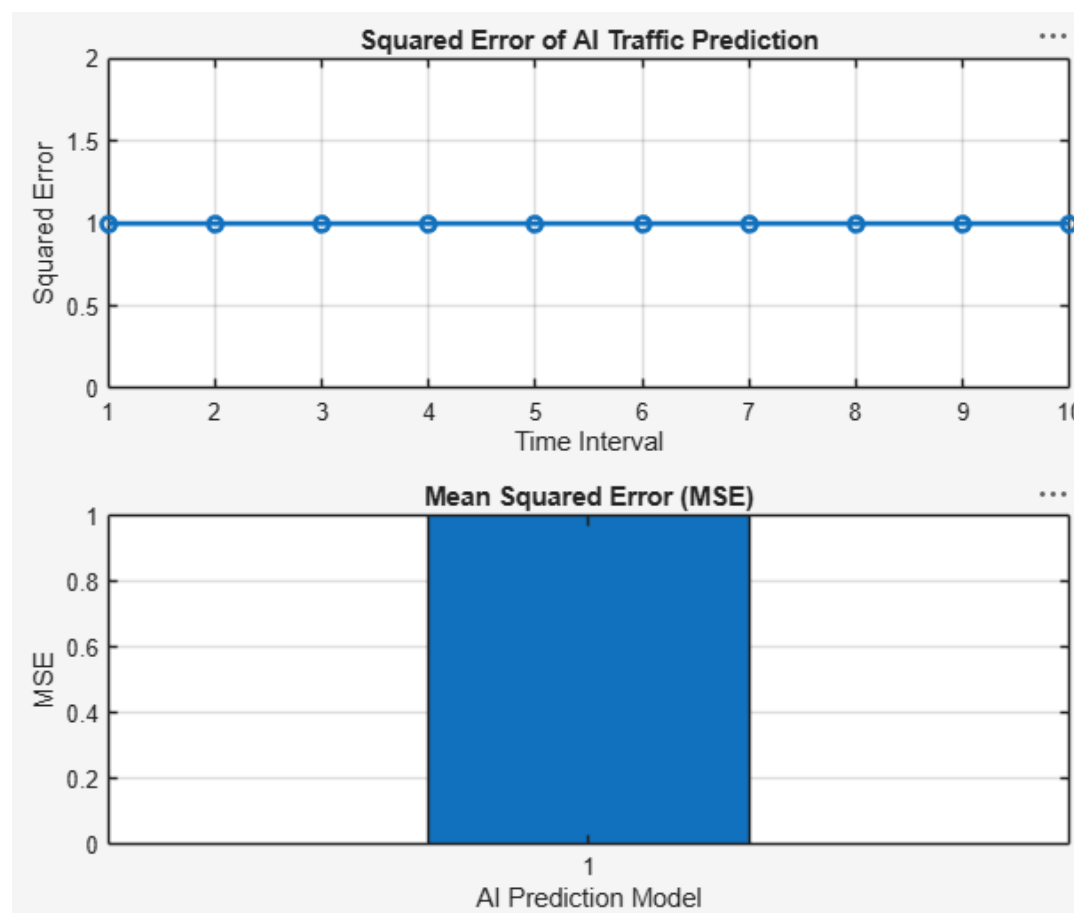
/MATLAB Drive/slotb15.m

```
1      clc;
2      clear;
3      close all;
4      % Time samples
5      t = 1:10;
6      % Prediction Accuracy (%)
7      Accuracy = [90 91 92 93 94 95 96 97 98 99];
8      % Actual Network Traffic (Mbps)
9      Traffic = [45 50 55 60 65 70 75 80 85 90];
10     figure
11     % ----- Graph 1 -----
12     subplot(2,1,1)
13     plot(t,Accuracy,'-o','LineWidth',2)
14     title('AI Prediction Accuracy over Time')
15     xlabel('Time Interval')
16     ylabel('Prediction Accuracy (%)')
17     grid on
18     % ----- Graph 2 -----
19     subplot(2,1,2)
20     bar(Traffic)
21     title('Network Traffic Samples Used for AI Prediction')
22     xlabel('Traffic Sample')
23     ylabel('Traffic (Mbps)')
24     grid on
```



Objective 2: Matlab Code and Output

```
26 % Time intervals
27 t = 1:10;
28 % Actual and Predicted Traffic (Mbps)
29 Actual = [50 55 60 62 68 70 75 80 85 90];
30 Predicted = [49 54 61 63 67 71 74 81 84 89];
31 % Squared Error
32 SqError = (Actual - Predicted).^2;
33 % Mean Squared Error
34 MSE = mean(SqError);
35 disp(['Mean Squared Error (MSE) = ', num2str(MSE)])
36 figure
37 % ----- Graph 1 -----
38 subplot(2,1,1)
39 plot(t,SqError,'-o','LineWidth',2)
40 title('Squared Error of AI Traffic Prediction')
41 xlabel('Time Interval')
42 ylabel('Squared Error')
43 grid on
44 % ----- Graph 2 -----
45 subplot(2,1,2)
46 bar([MSE])
47 title('Mean Squared Error (MSE)')
48 xlabel('AI Prediction Model')
49 ylabel('MSE')
50 grid on
```



Objective 3: Matlab Code and Output

```
52 % Time intervals
53 t = 1:10;
54 % Network Throughput (Mbps)
55 Throughput = [120 135 150 165 180 195 210 225 240 255];
56 % Network Utilization (%)
57 Utilization = [60 65 70 74 78 82 86 90 94 98];
58 figure
59 % ----- Graph 1 -----
60 subplot(2,1,1)
61 plot(t,Throughput,'-o','LineWidth',2)
62 title('AI-Based Network Throughput')
63 xlabel('Time Interval')
64 ylabel('Throughput (Mbps)')
65 grid on
66 % ----- Graph 2 -----
67 subplot(2,1,2)
68 plot(Throughput,Utilization,'-s','LineWidth',2)
69 title('Throughput vs Network Utilization')
70 xlabel('Network Throughput (Mbps)')
71 ylabel('Network Utilization (%)')
72 grid on
73 disp('Time Throughput(Mbps) Utilization(%)')
74 disp([t' Throughput' Utilization'])
```

